Project Design Phase-II
Data Flow Diagram & User Stories

Date	26 October 2026
Team ID	Gyanvi Jain
Project Name	Formula 1 Race Intelligence Dashboard
Maximum Marks	4 Marks

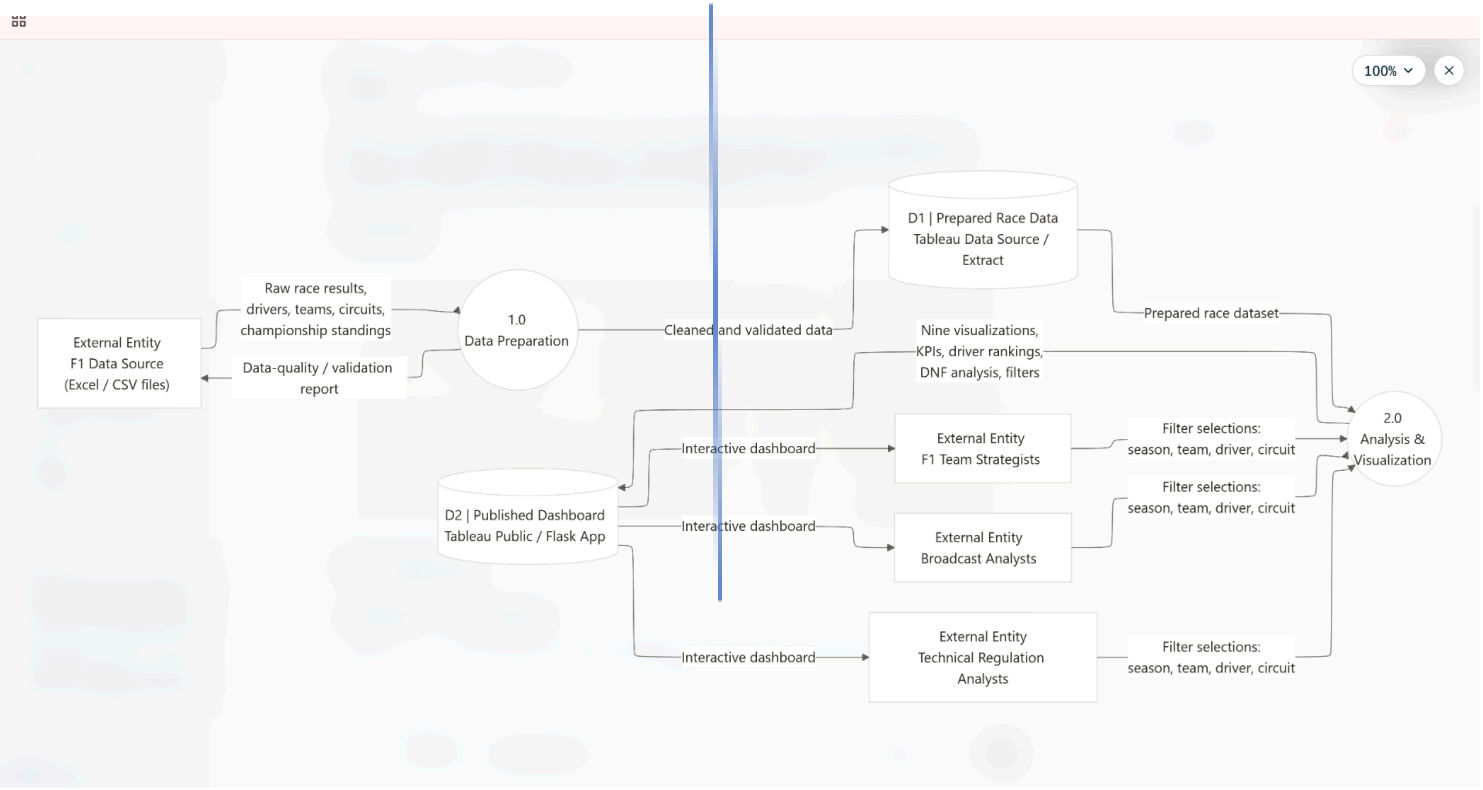
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web User)	Dashboard Access	USN-1	As a user, I can access the Formula 1 Race Intelligence Dashboard through a web browser.	The dashboard loads successfully and displays all available visualizations.	High	Sprint-1
Customer (Web User)	KPI Overview	USN-2	As a user, I can view key statistics about teams, seasons, race results, and circuits.	Four KPI cards display the relevant summary values.	High	Sprint-1
Customer (Web User)	Driver Performance	USN-3	As a user, I can view and compare driver performance using the leaderboard.	Driver rankings and cumulative points are displayed clearly.	High	Sprint-1

Customer (Web User)	Interactive Filters	USN-4	As a user, I can filter race data by DNF reason and year.	Selecting a filter updates the relevant dashboard visualizations.	High	Sprint-1
Customer (Web User)	Top N Parameter	USN-5	As a user, I can select how many top drivers appear on the leaderboard.	The leaderboard displays the number of drivers specified by the parameter.	High	Sprint-2
Customer (Web User)	Team Performance	USN-6	As a user, I can compare team performance and championship points across seasons.	Team rankings and championship performance are visualized for comparison.	High	Sprint-2
Customer (Web User)	Tyre Strategy Analysis	USN-7	As a user, I can explore tyre compound usage across race data.	The dashboard displays the distribution of tyre compounds.	Medium	Sprint-2
Customer (Web User)	Geographic Analysis	USN-8	As a user, I can explore driver distribution and country-wise race participation.	The geographic map and country-wise visualization display the available data.	Medium	Sprint-2

Customer (Web User)	Podium Analysis	USN-9	As a user, I can explore podium achievements by driver and team.	The packed bubble chart presents podium achievement data.	Medium	Sprint-2
F1 Team Performance Strategist	Performance Analysis	USN-10	As a strategist, I can analyse driver rankings, team performance, and championship trends.	The dashboard provides the relevant driver and team visualizations for analysis.	High	Sprint-2
Motorsport Broadcast Analyst	Data Retrieval	USN-11	As a broadcast analyst, I can quickly retrieve driver rankings, country participation, and podium statistics.	The relevant charts are accessible and can be explored for reporting purposes.	High	Sprint-2
Technical Regulation Analyst	Reliability Analysis	USN-12	As a technical analyst, I can investigate DNF patterns by failure reason.	The Driver Reliability Analysis chart supports exploration of available DNF categories.	High	Sprint-2
Administrator / Data Maintainer	Data Preparation	USN-13	As a data maintainer, I can prepare and validate race data before visualization.	Missing values and data types are checked, and required joins and calculations are applied.	High	Sprint-1

Administrator / Data Maintainer	Dashboard Publishing	USN-14	As a data maintainer, I can publish the dashboard for users to access online.	The published dashboard is accessible through its shareable Tableau Public URL.	High	Sprint-2